

C670

Concepts in Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher planning an astronomy unit notes that the Earth-Sun distance is about 1.5×10^{11} m and Earth's diameter is about 1.3×10^7 m. Light takes about 8 minutes to make the trip. Roughly how many Earths lined up would span the Earth-Sun distance?

- A. About 10,000
- B. About 2×10^{18}
- C. About 4
- D. About 480

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A millimeter is 10^{-3} m and a nanometer is 10^{-9} m. How many times larger is a millimeter than a nanometer?

- A. 10^{-6}
- B. 6
- C. 10^6
- D. 10^{-12}

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Students in a class of 28 measure a nature trail as 2.5 km long. The trail has 14 marked stops. How long is it in centimeters?

- A. 25,000 cm
- B. 2,500 cm
- C. 0.000025 cm
- D. 250,000 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A teacher who has run the science fair for six years lists four events for students to sort. Which one is a chemical change?

- A. An iron nail rusting in damp air
- B. Sugar dissolving in tea
- C. Ice melting on a counter
- D. Water boiling, because bubbles of gas form

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A chemistry teacher shows an atom from a seawater sample collected at a coastal field station. It has 17 protons, 18 neutrons, and 17 electrons. The sample was stored for three weeks. What is its mass number?

- A. 35
- B. 17
- C. 18
- D. 52

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Iron-56 has atomic number 26. How many neutrons does it have?

- A. 82
- B. 26
- C. 30
- D. 56

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A metal block cut into a cube shape by a local machinist has a mass of 60 g and a volume of 20 cm³. It came from a scrap bin behind the shop. What is its density?

- A. 0.33 g/cm³
- B. 1,200 g/cm³
- C. 3 g/cm³
- D. It depends on the cube shape.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Aluminum has a density of 2.7 g/cm³. What is the mass of a 10 cm³ piece?

- A. 12.7 g
- B. 27 g
- C. 2.7 g
- D. 0.27 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a unit on materials, a teacher heats magnesium ribbon from a 25-gram roll until it burns with a bright white light and forms magnesium oxide. Students wear goggles. What type of bonding holds magnesium oxide together?

- A. Covalent
- B. Hydrogen bonding
- C. Metallic
- D. Ionic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher writing a quiz for 90 students asks for the name of MgCl₂. The quiz is worth 20 points. Which name is correct?

- A. Magnesium dichloride
- B. Magnesium chloride
- C. Dichloride magnesium
- D. Magnesium chlorine

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Students heat a beaker of water on a hot plate bought last fall and record the temperature every minute for 10 minutes. The room is 21°C. How should they set up the graph?

- A. Room temperature on the y-axis
- B. Either way; axis choice never matters.
- C. Temperature on the x-axis, time on the y-axis
- D. Time on the x-axis, temperature on the y-axis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A graph of mass (g, y-axis) against volume (cm^3 , x-axis) for samples of one metal is a straight line through (0, 0) and (10, 27). What does the slope represent?

- A.** The mass of the metal, 27 g
- B.** The metal's density, 2.7 g/cm^3
- C.** Proof that adding volume causes mass
- D.** About $0.37 \text{ cm}^3/\text{g}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher ~~planning an astronomy unit~~ notes that the Earth-Sun distance is about 1.5×10^{11} m and Earth's diameter is about 1.3×10^7 m. ~~Light takes about 8 minutes to make the trip.~~ Roughly how many Earths lined up would span the Earth-Sun distance?

Target word: ratio of two magnitudes **Decides:** Divide: $10^{11} / 10^7 = 10^4$, with $1.5 / 1.3 \approx 1$.

A	Correct	$1.5 \times 10^{11} / 1.3 \times 10^7 \approx 1.2 \times 10^4$.
B	Inverted operation	Multiplies the two numbers instead of dividing.
C	Dropped element	Reports the exponent difference instead of 10 to that power.
D	Camouflage	Converts the 8-minute light time to seconds; travel time does not decide the ratio.

2. Answer C Concept 1 • transfer

A millimeter is 10^{-3} m and a nanometer is 10^{-9} m. How many times larger is a millimeter than a nanometer?

Target word: ratio of negative powers **Decides:** $10^{-3} / 10^{-9} = 10^6$.

A	Inverted operation	Divides the smaller by the larger.
B	Dropped element	Gives the exponent, not the ratio.
C	Correct	Subtracting exponents: $-3 - (-9) = 6$.
D	Confusion set	Adds the exponents, which is multiplication.

3. Answer D Concept 2

Students ~~in a class of 28~~ measure a nature trail as 2.5 km long. ~~The trail has 14 marked stops.~~ How long is it in centimeters?

Target word: km to cm **Decides:** $2.5 \text{ km} \times 1,000 \text{ m/km} \times 100 \text{ cm/m}$.

A	Wrong stem number	Uses 10 cm per meter instead of 100.
B	Dropped element	Converts only to meters, then relabels.
C	Inverted operation	Divides by the factors instead of multiplying.
D	Correct	$2.5 \times 1,000 \times 100 = 250,000 \text{ cm}$.

4. Answer A Concept 3

A teacher ~~who has run the science fair for six years~~ lists four events for students to sort. Which one is a chemical change?

Target word: identify a chemical change **Decides:** Rust is a new substance (iron oxide).

A	Correct	Iron and oxygen form a new compound.
B	Adjacent category	Dissolving mixes substances without changing their identity.
C	Confusion set	A change of state; still water.
D	Dropped element	The gas is still water; bubbles alone do not mean a new substance.

5. Answer A Concept 4

A chemistry teacher shows an atom from a seawater sample ~~collected at a coastal field station~~. It has 17 protons, 18 neutrons, and 17 electrons. ~~The sample was stored for three weeks.~~ What is its mass number?

Target word: mass number **Decides:** Mass number = protons + neutrons.

A	Correct	$17 + 18 = 35$ (chlorine-35).
B	Confusion set	That is the atomic number.
C	Wrong stem number	The neutron count alone.
D	Adjacent category	Adds electrons, which have negligible mass.

6. Answer C Concept 4 • transfer

Iron-56 has atomic number 26. How many neutrons does it have?

Target word: solve for neutrons **Decides:** Neutrons = $56 - 26$.

A	Inverted operation	Adds instead of subtracting.
B	Confusion set	That is the proton count.
C	Correct	$56 - 26 = 30$.
D	Wrong stem number	That is the mass number.

7. Answer C Concept 5

A metal block ~~cut into a cube shape by a local machinist~~ has a mass of 60 g and a volume of 20 cm^3 . ~~It came from a scrap bin behind the shop.~~ What is its density?

Target word: density **Decides:** Density = mass / volume.

A	Inverted operation	Divides volume by mass.
B	Inverted operation	Multiplies mass by volume.
C	Correct	$60 / 20 = 3 \text{ g/cm}^3$.
D	Camouflage	Density does not depend on shape.

8. Answer B Concept 5 • transfer

Aluminum has a density of 2.7 g/cm^3 . What is the mass of a 10 cm^3 piece?

Target word: solve for mass **Decides:** Mass = density $\times$ volume.

A	Adjacent category	Adds the two numbers.
B	Correct	$2.7 \times 10 = 27 \text{ g}$.
C	Absolute language	Treats density as if it were the mass of any sample.
D	Inverted operation	Divides density by volume.

9. Answer D Concept 6

In a unit on materials, a teacher heats magnesium ribbon from a 25-gram roll until it burns with a bright white light and forms magnesium oxide. ~~Students wear goggles.~~ What type of bonding holds magnesium oxide together?

Target word: bond type **Decides:** Magnesium is a metal; oxygen is a nonmetal.

A	Confusion set	Covalent bonds form between nonmetals.
B	Adjacent category	A weak attraction between molecules, not the bond within this compound.
C	Dropped element	Ignores oxygen; metallic bonding is metal with metal.
D	Correct	Mg gives two electrons to O, forming Mg^{2+} and O^{2-} .

10. Answer B Concept 7

A teacher writing a quiz for 90 students asks for the name of MgCl_2 . ~~The quiz is worth 20 points.~~ Which name is correct?

Target word: ionic name **Decides:** Metal + nonmetal: no prefixes.

A	Confusion set	Uses the prefix system that applies only to molecular compounds.
B	Correct	Ionic names use the metal and the -ide ending without prefixes.
C	Inverted operation	Reverses the order; the cation is named first.
D	Dropped element	Omits the -ide ending for the anion.

11. Answer D Concept 8

Students heat a beaker of water on a hot plate ~~bought last fall~~ and record the temperature every minute for 10 minutes. ~~The room is 21°C.~~ How should they set up the graph?

Target word: axis assignment **Decides:** Time is controlled by the students (independent).

A	Camouflage	Room temperature is a constant condition, not the measured variable.
B	Absolute language	Convention matters for reading slope as rate.
C	Inverted operation	Places the dependent variable on the x-axis.
D	Correct	The independent variable goes on the x-axis.

12. Answer B Concept 8 · transfer

A graph of mass (g, y-axis) against volume (cm^3 , x-axis) for samples of one metal is a straight line through (0, 0) and (10, 27). What does the slope represent?

Target word: meaning of slope **Decides:** Slope = $\Delta\text{mass} / \Delta\text{volume}$.

A	Wrong stem number	A single point's y-value, not the slope.
B	Correct	$27 \text{ g} / 10 \text{ cm}^3 = 2.7 \text{ g/cm}^3$.
C	Adjacent category	A slope describes a relationship, not cause.
D	Inverted operation	Divides x by y.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Scientific notation and orders of magnitude	To compare sizes, divide: subtract exponents of ten and keep the result as a power of ten.	Reporting the difference in exponents as the answer itself, or multiplying when comparing.	Two very large or very small numbers and “how many times.”
2	Unit conversion by dimensional analysis	Multiply by conversion factors arranged so unwanted units cancel, one step at a time.	Skipping a step (km straight to cm with one factor) or flipping a factor.	A measurement in one unit and a request for another.
3	Physical vs. chemical change	A chemical change makes a new substance; a physical change alters form or state but not identity.	Treating bubbles or dissolving as proof of a chemical change.	Rusting, burning, or cooking vs. melting, boiling, or dissolving.
4	Atomic number and mass number	Atomic number = protons; mass number = protons + neutrons; neutrons = mass number – atomic number.	Mixing up atomic number and mass number, or counting electrons in the mass.	Counts of protons, neutrons, or a name like carbon-14.
5	Density	Density = mass / volume; it is an intensive property that does not depend on sample size or shape.	Multiplying mass and volume, or thinking a bigger piece is denser.	A mass and a volume, or a sink/float question.
6	Ionic, covalent, and metallic bonds	Metal + nonmetal forms ionic bonds (electron transfer); nonmetal + nonmetal forms covalent bonds (sharing); metal + metal forms metallic bonds.	Calling any compound with a metal “metallic.”	The two elements in the compound.
7	Naming binary compounds	Ionic compounds name the metal then the nonmetal with -ide and no prefixes; molecular (covalent) compounds use prefixes like di- and tri-.	Using prefixes for ionic compounds.	A formula with a metal (ionic) or two nonmetals (molecular).
8	Reading and graphing data	The independent variable goes on the x-axis; the slope of a line is the change in y per unit change in x and carries units.	Reading a single point as the slope, or reading slope as cause.	A graph or table and a question about slope or axes.

Course scope (public description): Concepts in Science for graduates provides already-licensed teachers seeking an additional license or endorsement in science education with an introduction to essential science themes present within and across all science disciplines, including chemistry, physics, biology, and the geosciences. These themes include comprehending the magnitude of the physical and natural world, analyzing and converting measurements, understanding the basic nature and behavior of matter and energy, examining atomic structure, identifying and naming basic types of chemical bonds, and analyzing and interpreting scientific data. Concepts in Science provides a solid foundation for future, in-depth, scientific studies and should be taken prior to any other science content course. There are no prerequisites for this course.